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Transdisciplinary Methodology in Home Economics

Sue L. T. McGregor

Mount Saint Vincent University

Abstract

A transdisciplinary methodology better ensures that home economics can take the lead in ensuring that higher education disciplines (including home economics), government agencies, the corporate sector, and members of civil society can share perspectives and power to solve increasingly complex problems of humanity. To that end, this paper profiles home economics as comprising disciplinary work, interdisciplinary work and transdisciplinary work. It envisions home economics practice through the lens of transdisciplinarity as a methodology, with four axioms: (a) multiple Levels of Reality (ontology), (b) the Logic of the Included Middle, (c) knowledge as emergent complexity (epistemology), and (d) integral values constellation (axiology).

Keywords: transdisciplinarity, interdisciplinarity, methodology, home economics, complex human problems

Home economics is "the only critical, profound, and global profession that addresses the actual spectrum of nurturing basic human universals" (Davis, 1993 p. 32 (see also Davis, 2008)). To live up to the laudable goal of addressing problems of humanity, home economics could consider transitioning from an interdisciplinary to a transdisciplinary approach to practice. The latter involves intellectual, border-crossing work (Horlick-Jones & Sime, 2004) for the good of humanity (Brown, 1993). To facilitate a conversation about what this practice might look like in home economics, this paper shares ideas about transdisciplinarity as a methodology, an idea worth considering for home economics.

For a century, the profession has framed itself as interdisciplinary (Brown, 1993). Interdisciplinarians strive to assemble different disciplines around particular themes without any commitment to change the boundaries and relations among the disciplines (Fairclough, 2005). Home economics excels at this task. However, the unintended results often are fragmented knowledge, insular islands of specialized expertise, and too few connections with other disciplines that also are interested in problems faced by humanity (Brown, 1993; von Schweitzer, 2006). Home economics' unique contribution is its focus on the home and family. When home economics professionals embrace the idea of crisscrossing back and forth across its disciplinary boundary and those of other disciplines, with the intent of removing the boundaries to create a space for intellectual synergy, they will augment their ability to meaningfully contribute to the solution of the world's problems.

Acknowledging home economics' traditional interdisciplinary posture, McGregor (2010b) shared a history of the idea of transdisciplinarity in home economics. She recounted the

leading-edge contributions of Christine Daniels (1980) from Wales and Marjorie Brown (1993) from the United States. Daniels believed that transdisciplinarity was a way to organize home economics curriculum in higher education. Brown believed transdisciplinarity was both (a) an overarching conceptual framework and (b) a context for the unity of world views. She ardently maintained that by using a transdisciplinary framework, interdisciplinarian home economists would be better able to study more relevant social problems than if they remained constricted by their narrow disciplinary boundary and siloed specializations.

Daniels (1980) and Brown (1993) started the profession on its journey towards transdisciplinarity, using then-contemporary understandings of the concept. Daniels relied on Jantsch's (1972a,b) and Brown relied on Kocklemans' (1979) conceptualizations of transdisciplinarity. McGregor (2010b) observed that as the concept of transdisciplinarity evolved during the seventies and eighties so too did home economists' interpretations of transdisciplinary-informed practice. Indeed, Daniels and Brown tendered quite different ideas of what transdisciplinary in home economics might look like. They both provided very valuable insights for the profession, nudging us further away from our historical reliance on interdisciplinarity.

In her discussion of Daniels' (1980) and Brown's (1993) contributions, McGregor (2010b) asserted that, by remaining cognizant of the continually evolving notion of transdisciplinarity, the profession could become a key player in the global movement for integral, boundary-crossing approaches to address the magnitude and complexity of problems faced by humanity. Following in Brown's and Daniels' footsteps, she continues to champion the idea of transdisciplinarity and home economics; however, McGregor draws on new iterations of transdisciplinarity that have evolved since Daniels' and Brown's contributions. In particular, she employs notions of transdisciplinarity as methodology, a perspective that originated at the first world congress on transdisciplinarity (cf. Nicolescu, 2002). This meeting was held in 1994 in Portugal, one year after Brown's work was published and 14 years after Daniels' paper.

For clarification, earlier notions of transdisciplinarity (TD) assumed it was a hyper form of interdisciplinarity (Jantsch, 1972a,b; Kocklemans,1979). New scholars of transdisciplinarity perceive it as more; they envision it as an integrated combination of: (a) disciplinary work, (b) scholarship between and among disciplines (interdisciplinarity), and (c) knowledge generation beyond academic disciplines and across sectors external to the university (at the interface between the academy and civil society) (McGregor & Volckmann, 2010) (see Figure 1). This approach is the focus of this paper.

Home Economics Disciplinary Work

Home economics concurrently evolved as both a discipline and a profession. The former entailed the development of university units, higher education curricula, library holdings, and academic journals and gatherings. Disciplines are understood to be areas of academic study that are comparatively self-contained and isolated domains of learning, which possess their own community of experts (Nissani, 1997). On the other hand, a profession involves a responsibility to serve the public, has a complex, evolving body of knowledge (drawing from disciplinary scholarship), has standards of admission, requires certification or licensing, and

has a need for public confidence (Brown, 1993). Professions require professional associations, accreditation and/or licensing, ongoing professional development and in-servicing, and codes of ethics and practice, even regulatory legislation.

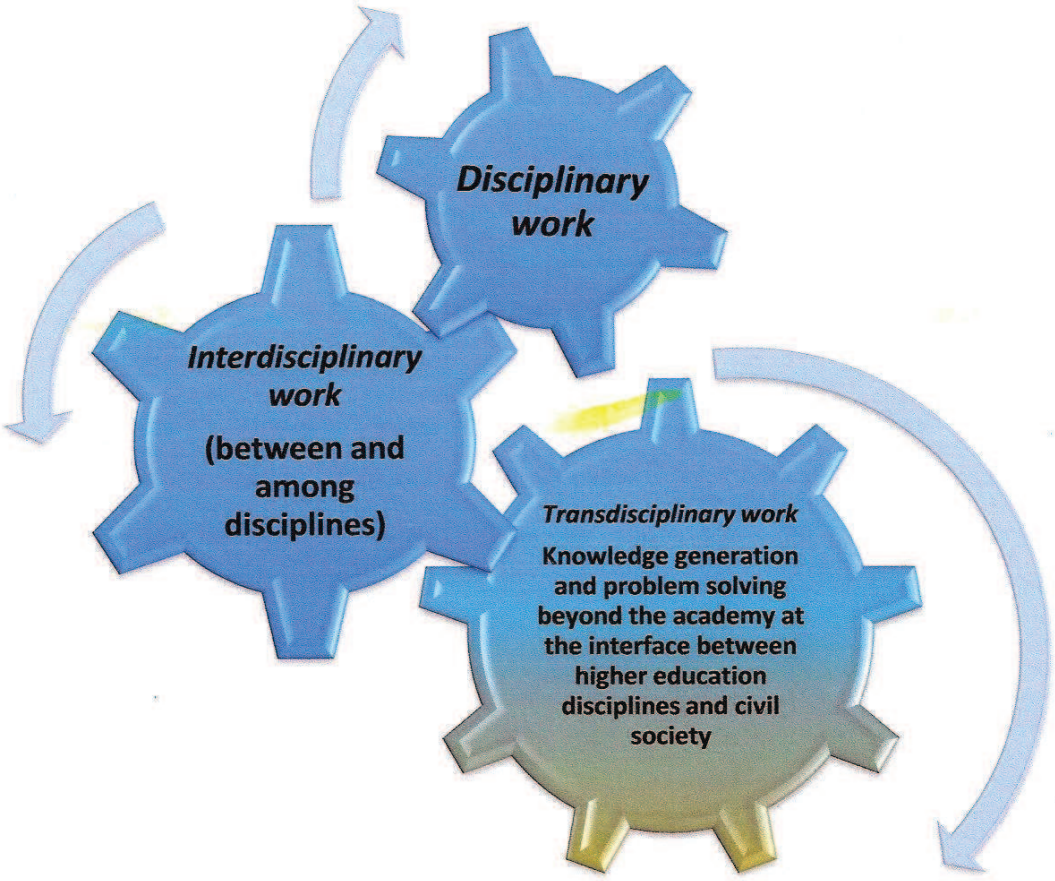


Figure 1 Contemporary notion of transdisciplinarity - integrated combination of three forms of scholarship (work) and problem solving

In more detail, the academic discipline of home economics has created (and continues to create) its own disciplinary knowledge base, to serve both a generalist and a specialist approach. The former prepares mission-oriented professionals to assist families in complex problem solving. It often involves added attention to the stated mission, values and philosophical core of the profession. The specialist approach prepares professionals for specific career paths, with the focus on the mission less clear (Butler, Ade-Ridder, Rudge & Senbach, 1987). Brown (1993) further clarified that a generalist is prepared to integrate knowledge from a variety of disciplines while a specialist studies within existing sub-divisions of an established discipline (in our case, food, clothing, shelter, family, consumption et cetera). Turkki (2006) proposed a new kind of specialization within home economics. We

would become integral specialists with expertise that integrates, links bridges, sees connections, looks for patterns, coordinates and communicates.

As well, within the last decade, some professional associations have focused attention on clarifying the profession's body of knowledge (BOK) (e.g., Nickols et al., 2009). A BOK is a term used to represent the complete set of agreed-to concepts, terms, principles and activities that make up a professional domain, as defined and advocated by the relevant professional association. BOKs contain the main domains of learning and respective high-level topics or knowledge areas, as developed within the academic discipline. Ideally, there is a mutually beneficial relationship between the discipline and the profession (McGregor & MacCleave, 2007), one which ensures productive and meaningful disciplinary work that advances the discipline and serves the profession.

Home Economics Interdisciplinary Work

Brown (1993) clarified that interdisciplinary work involves the integration of knowledge and modes of reasoning and logics from different disciplines into new patterns of thought and practice (taken to mean empirical logic, interpretive logic (meanings) and critical logic (judgements about power infrastructures)). From an interdisciplinary stance, problems faced by individuals and families are perceived and dealt with through inquiry in research and in education over several disciplines (including some combination of the natural, human, social, and administrative sciences). Daniels (1980) and McGregor (2008, 2010c) explained that home economics higher education programs are conventionally structured so that students take courses from other disciplines. Their minds are trained to tease out content, theory and principles from aligned disciplines and then to draw on the synergy created when connections are made between these insights and home economics' mission and philosophy. The intended result is interdisciplinary-informed practice that is focused on individual and family needs and functions.

It bears repeating that, despite this deeply valuable approach to academic socialization into the profession, interdisciplinarians strive to assemble different disciplines around particular themes without any commitment to change the boundaries and relations among the disciplines (Fairclough, 2005) or between the discipline of home economics and other disciplines. In our case, architects of higher education programs design curricula that expose students to an array of disciplines, with the intent to create a home economics-focused disciplinary knowledge base. Over time, this knowledge base emerged as a result of the generation of fresh knowledge produced through the integration of knowledge from a collection of different disciplines (Brown, 1993). Done well, interdisciplinary approaches help home economics students create patterns for interpreting the world through a multitude of disciplinary lens, while remaining focused on family well-being and quality of life.

Brown (1993) and others also characterized home economics as multi-disciplinary. Interdisciplinary work differs from multi-disciplinary studies in that the latter does not strive to integrate knowledge into new patterns. Both approaches are natural first steps for solving complex social problems, but they are not enough (Brown, 1993; Lattanzi, 1998). Human problems do not exist in isolated discipline-oriented packets; hence, solutions to such complex problems cannot be found in disciplinary, even multi- or inter-disciplinary,

approaches. To that end, Brown asked us to use the transdisciplinary framework to re-examine the purpose historically claimed for home economics, both its meaning and its justification.

Home Economics Transdisciplinary Work

Brown (1993) portrayed transdisciplinarity in home economics as an overall framework for the synthesis of knowledge that unites and then transcends the disciplines because it focuses on work done in various fields by people concerned with "the human problem" (p.296). She astutely explained that this knowledge synthesis must take place in the mind of each home economist before a collective synthesis can take place within the discipline. This synthesis cannot occur if each home economist seeks comfort within his or her specialist-trained mind, limiting his or her perspective concerning the human problem. Home economists engaged in this transdisciplinary process would have to move beyond their views of the world informed by their respective discipline (Brown). Even if it is interdisciplinary in its philosophy, home economics is not the only way to see the world.

Transdisciplinarity also significantly deepens the range of problems that home economists would consider. Historically, we have focused on the quality of life and well-being of individuals and families. This characterization of our raison d'etre fails to "clarify the set of human problems appropriate for home economics" (Brown, 1993, p.255). Transdisciplinarity would have home economists "go to the root of human problems ...rather than to treat the manifest symptoms [thereby] making us more critically reflective about social realities and their influence on the human condition rather than blindly accepting existing social conditions [characterized as well-being and quality of life]..." (p.268, emphasis added). Present day scholars (e.g., Nicolescu, 2001) agree that the set of human problems mentioned by Brown encompasses: the human condition, issues that have global implications, human freedom and justice, self-determinism, harmonious access to and distribution of resources, human and bio sustainability, power relationships, human aggression, human development, human empowerment and potential, and the import of ideologies, worldviews and paradigms. This notion of what constitutes a problem is a far cry from home economics' conventional rhetoric of specialized issues pertaining to food, clothing, shelter, family relations and individual development, and consumption-related scenarios.

As mentioned before, Sue McGregor is carrying on Christine Daniels' and Marjorie Brown's work by continuing to champion transdisciplinarity and home economics. In her paper for KON's Human Sciences Working Paper Series, McGregor (2004) distinguished transdisciplinarity from other forms of disciplinarity with which home economists are familiar. She then used metaphors to illustrate the mind shift necessary to move home economists from multi- and inter-disciplinarity to transdisciplinarity. She concluded by tendering a transdisciplinary self-orientation tool to help individual practitioners determine their readiness for this new approach to practice. More recently, her writings affirm her belief that transdisciplinarity is a methodology in its own right,1 akin to empirical, interpretive and critical methodologies (McGregor, 2005, 2007a, 2009a, 2010a; Torkar & McGregor, 2012). This paper now turns to this current approach to transdisciplinarity, and what it might mean for home economics practice.

Transdisciplinarity as a Methodology

Brown (1993) exclaimed that it is not enough for home economists to solve the set of human problems using an integrated package of technical information from several disciplines (the logic of empirical scholarship). Such problems also concern human values (interpretive logic) and the need to judge and critique global resource distribution and resultant power infrastructures (critical logic). She asked us to apply a combination of all three methodologies to solve the problems of humanity - empirical, interpretive and critical. McGregor and Murnane (2010) provided a detailed discussion of how these three methodologies differ.

Contemporary approaches to transdisciplinarity actually conceive it as a fourth methodology in its own right, with its own notions of what counts as knowledge, logic, reality and the role of values (four axioms, with axiom meaning self-evident rules, principles and laws). Nicolescu (2002, 2008b) is the leading advocate of the transdisciplinary (TD) methodology. He tendered three pillars (axioms) of transdisciplinarity: (a) multiple Levels of Reality (ontology), (b) the Logic of the Included Middle, and (c) knowledge as emergent complexity (epistemology). McGregor (2009a) and Cicovacki (2004, 2009) added a fourth pillar of values (axiology), which McGregor labelled integral values constellation (see Figure 2).

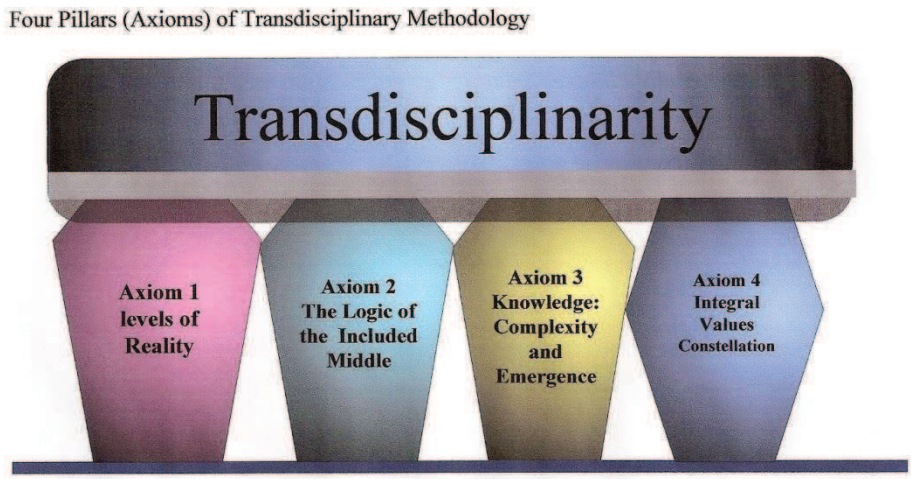


Figure 2 Four pillars (axioms) of transdisciplinary methodology

It is important to note that the current TD methodology is deeply informed by the new sciences of quantum physics, chaos theory and living systems theory (Nicolescu, 2006). By comparison, most home economists have been socialized in higher education programs informed by Newtonian physics. The latter (known as Modern Science) assumes everything is separate, especially that science is separate from theology, philosophy and culture. It embraces fragmentation, dualities (this or that), and universal laws that apply to everything and everyone (with no concern for context). The world is viewed as black and white. The ideology of scientism assumes that only knowledge generated using the scientific method is

worthy knowledge. Objectivity becomes the extreme criterion for Truth and reality. Human beings become objects.

Within this scientific mind set, generations of home economists have become comfortable with being experts, with working within their own discipline, and especially with being specialists within home economics sub-disciplines (hyper-specialization). They have become adept at assuming that their picture of family reality is incomplete, made up of many separate parts. If they can just conduct enough experiments and develop enough theories about this singular reality, they will eventually build up a more complete picture (McGregor, 2009b). Nicolescu (2006) explained that the innovation of the new sciences radically changed what is acceptable as knowledge, reality, logic and the role of values. Home economics now has a methodology that enables it to renounce (at least downplay) scientism, fragmentation and hyper-specialization, a fervent call of Brown (1993). Each of these four pillars of the TD methodology is now discussed.

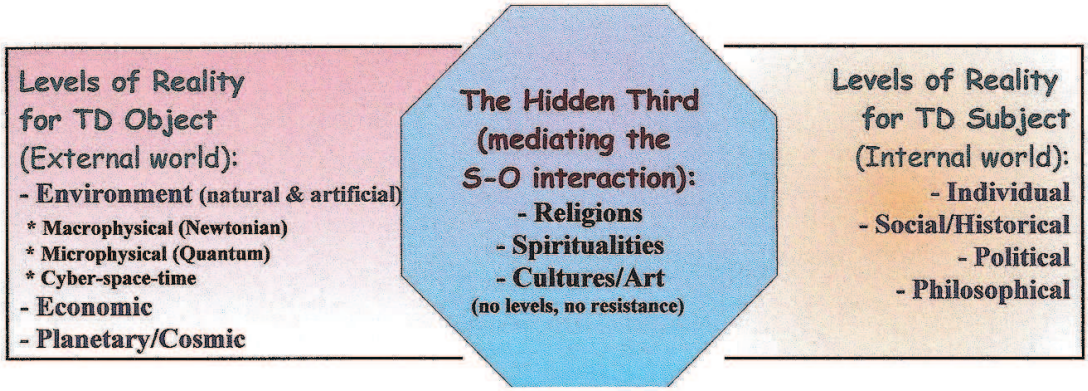
Transdisciplinary Reality (Ontology)

Ontology refers to the study of existence or reality. It deals with questions of what entities exist (or can be said to exist) and of how they can be grouped or sub-divided according to similarities or differences. Instead of one singular reality, transdisciplinarity accommodates 10 key realities (multiple levels of realities): the environments, economics, politics, culture and art, social and historical, individual and community, the planet and the cosmos. Each of these realities is characterized by its incompleteness; yet, together, in unity, these realities generate new, infinite knowledge (Nicolescu, 2006).

Some people are more aware of different levels of reality than are others (i.e., it is their speciality, their expertise, or their lived experience). For this reason, transdisciplinarity assumes that it is essential to seek multiple perspectives on any human problem (or set of human problems) because the intent is to integrate many levels of truth. There is no right or wrong, good or bad, yes or no answer; there is no black and white. Also, no one level of reality constitutes a privileged place from which individuals are able to understand all other levels of reality; instead, a level of reality is what it is because all of the other levels exist at the same time (Nicolescu, 2006).

From a TD perspective, home economists would respect the complex and dynamic relationships between three levels of reality: (a) the internal world of humans (comprising political, social, historical and individual realities); (b) the external world of humans (comprising environmental, economic, and cosmic/planetary realities); and, The Hidden Third. People's experiences, interpretations, descriptions, representations, images, and formulas meet on this third level. Three realities exist in this intuitive zone/level: culture and art, religions, and spiritualities (see Figure 3).

Levels of TD Transreality for Subject and Object, with Hidden, Mediating Zone
(I am grateful to Basarab Nicolescu for his personal contributions to my understanding of this concept)



In summary, the TD methodology assumes there are 10 different realities, organized along three levels. Together, they form TD ontology: the internal world of humans (with its flow of consciousness), the external world (with its flow of information), and their mediated interface (Nicolescu, 2006, 2008b). Home economists would no longer problem-solve from one level of expertise (e.g., just home economics, or just politics or just economics). They would expand their horizons to envision solving the set of human problems by employing a different notion of what counts as reality, and by accepting that these realities are intertwined.

Although this idea may feel similar to human ecology's concept of reciprocal relationships among levels of environments, it is different in that it deals with the mediated flow of inner consciousness (perceptions) and technical information from different realities leading to a meeting of the minds in a Zone of Non-Resistance. In this space, home economists would shed their resistance to truth informed by other realities, and join these realities to generate complex, TD knowledge (see Figure 4).

Transdisciplinary Logic

Transdisciplinarity asserts there are multiple levels of reality and that each level is governed by a different kind of logic (Nicolescu, 2008b). Logic is concerned with the habits of the mind that are acceptable for inference and reasoning when arguing one's position on an issue. In philosophy, logic is concerned with answering the question "How do we know what we know?" Logic (reasoning and inference) is used in most intellectual activity, and perceptions of what counts as logic underpin all intellectual activity. The tendency in higher education is to predicate curriculum on the Newtonian logic of exclusion, on reductionist-based scientific inquiry (Nicolescu, 2008b). The logic of exclusion assumes that the space between objects or people is empty, flat, static and void of life (like the space between the balls on a billiard table). In academic life (and home economics is no exception), this logic manifests as

separate departments, journals, library holdings, conferences and professional associations. It is also evident in the familiar intellectual actions of: deduction (cause and effect), linear thinking, reductionism (breaking things down into parts to understand the whole from which they come), and either/or approaches with no room for contradictions.

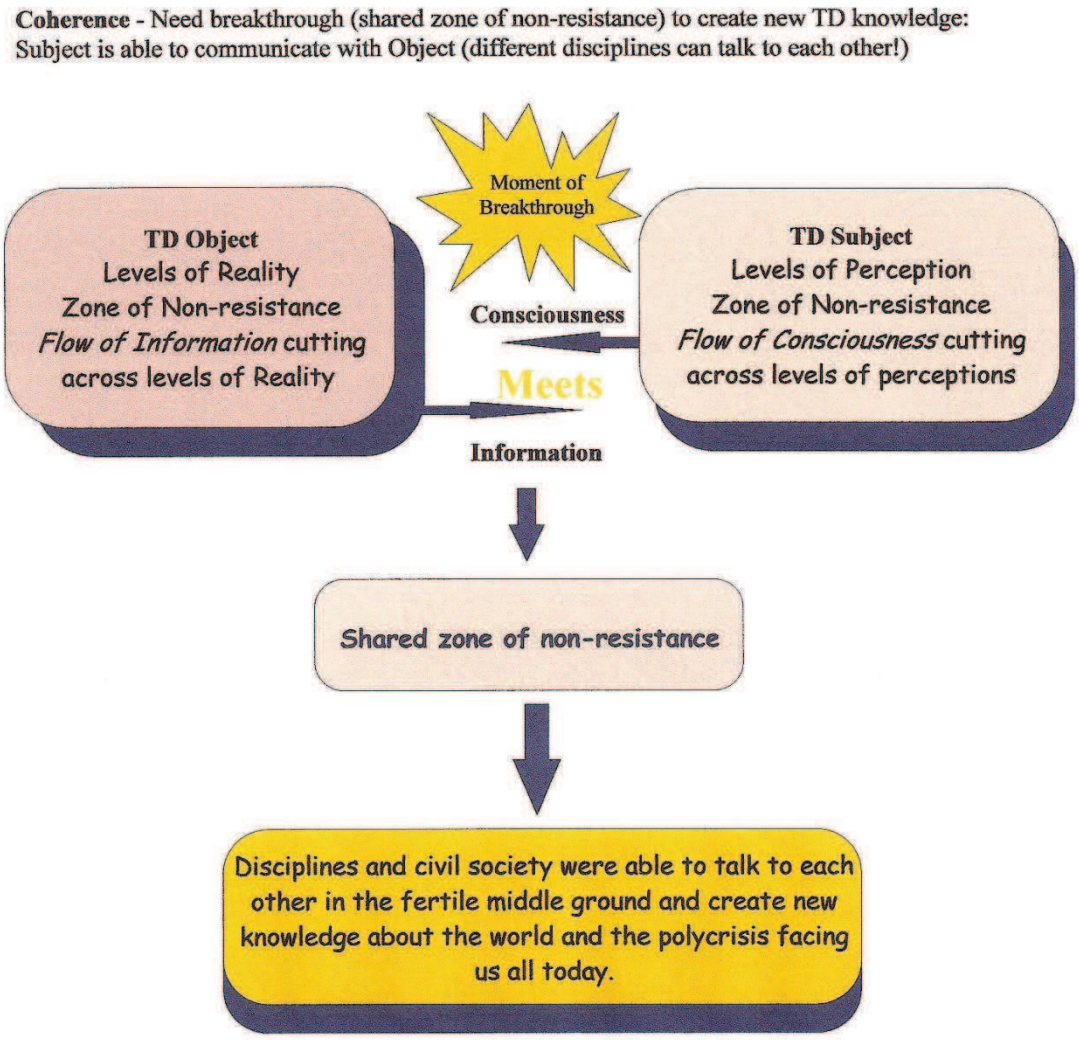


Figure 4 Transdisciplinary zone of non-resistance where actors can talk with each other

Home economists often employ the logic of exclusion, even though they characterize themselves as interdisciplinary. Interdisciplinarity builds bridges between disciplines so ideas can cross back and forth across temporary, coordinated borders. The assumption is that a bridge is needed to cross the deep chasm between siloed, fragmented fields of study and between hyper-specialized sub-disciplines within home economics (McGregor, 2004, 2006). In practice, the lack of a temporary bridge means there are many instances when people from different disciplines (even within the (sub)discipline(s) of home economics) cannot talk to

each other; hence, there can be no integration or generation of new knowledge (MacCleave, 2006).

In contrast to a logic of exclusion, a TD methodology embraces the Logic of the Included Middle. This inclusive logic enables people to imagine that the space between things is alive, dynamic, in flux, moving and perpetually changing. It is in this fertile middle space that transdisciplinary manifests itself. Transdisciplinarity has people stepping through the zones of non-resistance (away from one worldview or notion of reality towards others) onto a fertile, moving floor of the included middle, where, together, they generate new transdisciplinary intelligence and knowledge. The logic of the included middle would require home economists to create a space for dialogue and knowledge generation. In this space, attempts would be made to reconcile different logics for the sake of solving the set of human problems. Using the logic of the included middle to move through the 10 different types of reality (by making space for contradictions and discontinuities in realities), creates a permanent possibility for the evolution of new knowledge.

In more detail, quantum physics allows people to imagine many fuzzy-edged balls of knowledge and ways of knowing coming together. When all of these people and their ideas collide on the fluxing, fertile space, fusion occurs when separate bits of knowledge come together. These knowledge and perception/consciousness bits move faster when they are exposed to each other (McGregor, 2004). TD knowledge is created. The logic of the included middle allows people to bridge adjacent levels of reality, enabling meaning-making systems to orient themselves to each other (Nicolescu, 2008b). When people from different disciplines come in contact with each other and are motivated, an energizing force is generated - a synergy is created. This synergy leads to the generation of embodied knowledge created from the energy emanating from the intellectual fusion. McGregor (2009a) employed a metaphor of a lava lamp to explain this idea. As well, a sense of community and belonging is nurtured - a sense that individuals are part of something bigger than each of them. At the same time, there is a realization that everyone is a new and different person in each relationship(s) formed in the fertile middle. The strength and potentialities that emerge from these intellectual encounters are life-giving and transformative.

What is especially significant about the logic of the included middle is transdisciplinarity's focus on what happens at the interface between higher education (academic disciplines) and the rest of the world. In order for home economics scholars to become transdisciplinary, they would have to connect with members of civil society (as well as with the private and public sectors). Civil society refers to all groups outside government and the corporate sector (including the academy) such as: community groups, non-governmental organizations, labour unions, Indigenous Peoples' organizations, women's groups, charitable organizations, faith-based organizations, professional associations, foundations, and citizens (FAO, 2008). While inter means between, trans means to crisscross back and forth, to zigzag, to move beyond, to move through, to pass beyond the limits. By definition, this means home economists would reach beyond the walls of their academic discipline and sub-discipline(s) into the world of lived human experiences, interacting with actors from other sectors (McGregor, 2004, 2006).

Furthermore, a TD methodology would ask home economists to accept that what counts as knowledge has to expand beyond that which is generated using the scientific method to include the complex structure of understandings garnered through intricate webs of relations among individuals in the academy, the private and public sectors, and in civil society. Horlick-Jones and Sime (2004) coined the phrase border-work to refer to the intellectual work that occurs when people living on the borders of the academy (university disciplines) and civil society engage in knowledge generation to address complex problem solving.

Transdisciplinary Knowledge (Epistemology)

The third axiom is TD knowledge. Epistemology stems from the Greek word, epistēmē, meaning knowledge. It is a branch of philosophy concerned with what is knowledge, how is it acquired, what do people know and how do they come to know it. As with ontology and logic, transdisciplinarity strives for a different kind of epistemology than do the other three methodologies (see McGregor and Murnane, 2010). Transdisciplinary knowledge is based on cross-fertilization, embodiment, complexity and emergence.

A different understanding of these familiar words is required of home economists, especially complexity (versus complicated) and emergence. Transdisciplinarity draws its notion of complexity from the new sciences. From this perspective, there is a difference between a complicated problem and a complex problem. Although complicated and complex have the same root, they do not mean the same thing in a TD methodology. A complicated problem is characterized as hard to solve because it is intricate, tangled, knotty and detailed. It is one thing to untangle the strings of a complicated problem but quite another to re-weave them, along with new strings, into a new whole. The latter process is an inherent part of solving a complex problem because it features the concept of emergence.

McGregor (2010a,d, in press) provided an in-depth discussion of the TD concepts of complexity and emergence. Succinctly, transdisciplinarity holds that the process of emergence comes into being as people pass through the zone of non-resistance (accept that there are many realities) and enter the fertile middle ground to problem solve using the logic of inclusion. Emergence refers to novel qualities, properties, patterns and structures that appear from relatively simple interactions among people, qualities that did not exist when presented in isolation. These new qualities are layered in arrangements of increased complexity (Morin, 2005; Nicolescu, 2008a). Emergence means home economists would assume that the problem(s) they are addressing continually changes as people try to jointly solve it. The problem (for example, poverty) is a rich weave of societal structures and functions. This new weave of poverty (and people's understandings of poverty) keeps changing because new and coherent structures, patterns and properties emerge as a result of the interactions among people trying to address poverty while working within a web of changing relationships (on the included middle ground). Original perceptions of poverty are left behind or transformed as new understandings of the problem take shape and as the synergistic energy is generated during the intellectual border-work (McGregor, 2009a).

TD knowing assumes that everything is interconnected and interrelated in the world, which is far from being consistent and predictable. To that end, transdisciplinarity tries to understand the complex problems of the world (the big issues with global implications) rather than the

problems inside specific disciplines and sub-disciplines. To achieve this complex knowing, a new kind of knowledge is required, called TD knowledge. TD knowledge is created in the fertile middle ground, emerging from gradual cross-fertilization. The latter results from the iterant convergence of different actors and their fuzzy-edged balls of knowing, shaped by their respective expertise or disciplinary axiomatic (Nicolescu, 1997). The creation of TD knowledge depends on negative (maintain the status quo) and positive (deviate from the path) feedback to self-generate. Increasing deviation allows for transformation; hence, TD's notion of complexity implies that home economists would pair unlike ideas to generate intellectual deviations (Morin, 2005).

Nicolescu (1997) further suggested that generating TD knowledge requires people to become an apprentice of creativity, and to respect open unity and complex plurality. People have to figure out how they have been conditioned to learn, and they have to become comfortable with testing the foundations of their convictions. Also, TD knowledge emerges through the process of integration, understood to mean opening things up to all disciplines and to civil society-knowing so that something new can be created via synthesis and the harmonization of ideas and perspectives (different realities).

Cicovacki (2009) pointed out that theories of truth and knowing from the other methodologies are one-sided and inadequate for dealing with today's set of human problems. It is not that they completely miss the nature of truth and knowledge; rather, they capture only a few relevant aspects and disregard others. To address this issue, the TD methodology assumes that everything is complexus - woven into a web, where the focus is on the relationships (links) among things instead of on the things that are linked. The resultant knowledge that is created when higher education meets civil society is alive because the set of problems the knowledge addresses is alive, emerging from the life world (McGregor 2009b). When people accept the world and everything in it as dynamic, evolving and always in-formation, their knowledge, explanations and definitions gain nonpermanent status. TD knowledge is vibrant, alive and used in common rather than stagnant, hoarded and copyright protected (McGregor, 2004, 2006).

Finally, discussed in limited detail in this paper (see elsewhere in McGregor, 2010d), home economists also would come to welcome chaos, uncertainty, and tension, along with emergence and complexity. They would know that TD intellectual border work is going to lead to personal and disciplinary growth and evolution. Change creates chaos (a lack of regular, predictable arrangements, as defined by the new sciences). Home economists would learn to self-organize (reorganize) as they accept chaos and seek solutions to the apparent lack of order (the problems of humanity). This reorganization would lead to renewal (Wheatley, 1999). Home economists would not try to maintain the old order but would enter into trustful, sharing relationships with others who have the same vision and relevant information. Together, they would create a new world and generate creative solutions to complex, emergent human problems. Through rich processes and exchanges, multiple minds would interact and produce a complex knowledge containing its own reflexivity. This knowledge creation process is rich with value orientations.

Transdisciplinary Values (Axiology)

The final TD axiom deals with values. By its very nature, transdisciplinary dialogue will witness the inescapable value loading of every inference, every opinion; every line of conversation will face a potential clash of values, ethics and morals. The transdisciplinary methodology is very politicized, hence value laden. The fourth pillar of the TD methodology is thus concerned with axiology, the role of values and the researcher/change agent in the knowledge creation and problem solving process (Cicovacki, 2009; McGregor, 2009a).

The science of inquiry into human values (axiology) enables people to identify the internal valuing systems that influence their perceptions, decisions and actions (Schoof, 2009). Within transdisciplinary complex problem solving, thinking and action are intricately bound, necessitating a key focus on values. Indeed, Bazewicz (2000) affirmed that transdisciplinarity holds a holistic vision of the world, and is concerned with the local and global integration of values. Cicovacki (2009) argued that TD practitioners need to be able to deal with values and their systematic examination, even to develop a map of values. This values map would serve as a coordination system for the TD map of multiple realities that people are trying to assemble and navigate.

Working together in fluctuating, enriching and challenging relationships necessitates a concern for values, especially because complexity infers the need for more than a single expert's solution. In order to develop the necessary tolerance of different viewpoints, so people can stay engaged in conversations about the complex problems shaping the human condition, home economists would respect the role of axiology in transdisciplinarity. Küpers (2009) asserted that changes in value mixes are a key part of the rapidly changing global village, and that profound changes are taking place at all levels. He agreed with vanBreda (2007), that values often are the missing link in providing strategic solutions to key, global issues, which are informed by a collage of differing worldviews held by individuals, cultures, nations and regional and international groups. Transdisciplinarity needs value-steered processes to lead to multiple truths; its attendant TD knowledge production depends upon it (Burger, 2003).

Integral values constellations (Axiom 4) play a key role in successfully implementing the transdisciplinary methodology in home economics practice. The solutions for complex, emergent problems require partnerships with, and the knowledge of, experts from different scientific disciplines, from the private and public sector and from civil society. The capacity to solve complex problems is generated through these relationships, which are predicated on the generation and allocation of power, which is energy. This compilation and integration of different world views and value orientations may cause conflicts, which can result in power struggles. Whether or not the power (the energy force) generated when people work together to solve complex and emergent human problems is negative or positive depends upon the nature of the relationship(s). And, with power comes responsibility. Embracing this TD responsibility can be a challenge unless the constellation of values at play is respected, managed and lead.

Summary and Emergent Conclusions

In summary, if they employed a transdisciplinary methodology, home economists (a) would crisscross disciplinary and sub-disciplinary boundaries with the intent to change (or remove) the borders while integrating theories, policies and practices emanating from this disciplinary migration and integration. (b) They would recognize that leadership for home, families and humanity happens in the fertile middle ground within the academy and among higher education, civil society and other sectors. This integral leadership is informed by the logic of inclusion and multiple levels of reality. (c) Home economists would appreciate that TD knowledge is complex, emergent and embodied. They would find new respect for tension, chaos and disorder, (d) especially as they managed the value-laden transdisciplinary dialogue inherent in intellectual fusion. And, (e) they would integrate many realms of reality (multiple perspectives and logics) as they worked with other disciplines and members of civil society in intellectual border-work to address the context of 21st century humanity.

A transdisciplinary methodology better ensures that each of higher education, government agencies, the corporate sector, and members of civil society can strive to share their perspectives and power to solve increasingly complex problems of humanity. Home economics can assume a leading role in this process. Daniels (1980) and Brown (1993) started the profession on its journey towards transdisciplinarity, drawing on then-contemporary iterations of TD. This paper shared a third version of transdisciplinarity, informed by Basarab Nicolescu's conceptualizations of TD as a methodology in its own right. It explained how the current generation of TD might reshape home economics if members of the profession shifted paradigms. This shift would enable them to appreciate the merit of respecting TD as a fourth methodology, augmenting the familiar empirical, interpretive and critical methodologies.

This paper intentionally makes a strong value statement about the discipline and the profession. Brown (1993) claimed that home economists need to rationally examine any statements about what home economics can and should be, else we will lack the conviction and the will to act upon them. Believing that home economics should become convinced of the merit of acting on the TD methodology, this paper aimed to illustrate how transdisciplinarity can be a powerful way to reframe home economics. It could become a discipline that engages in disciplinary work, interdisciplinary work, and knowledge generation and problem solving at the interface between higher education and civil society - transdisciplinary work (see Figure 1). This transdisciplinary work would lead to a different, higher plane of inquiry involving the co-invention of new TD knowledge at the interface of disciplines, civil society and other sectors, and do so through joint problem solving, reorientation, perspective shifting, and value integration.

The TD methodology, with its focus on integral, complex and emergent problem solving, seems a natural fit with home economics' deep core. Home economics is a science of human life and existence, shaped by life within the home (Fusa, 2004). East (1979) asserted that home economics is a focus on the home in order to improve humanity. If we accept her insight into home economics' raison d'etre, we will need a new methodology with which to solve the set of human problems appropriate for home economics' focus and attention. In our contemporary times, that methodology is called transdisciplinarity.

Biography

Sue L. T. McGregor is a Canadian home economist at Mount Saint Vincent University, Canada. She is a Professor in the Faculty of Education. Her intellectual work pushes the boundaries of consumer studies and home economics philosophy and leadership towards integral, transdisciplinary and moral imperatives. She is Docent in Home Economics at the University of Helsinki, Marjorie Brown Distinguished Professor and a Kappa Omicron Nu Research Fellow. Affiliated with 20 professional journals, she is Associate Editor of three home economics journals. Sue has delivered 35 keynotes and invited talks in 11 countries and published over 120 peer-reviewed publications, 11 book chapters, and eight monographs. She published Transformative Practice in 2006. Consumer Moral Leadership was released in 2010. With Russ Volckmann, she co-published Transversity in 2011. Her professional and scholarly work is available at http://www.consultmcgregor.com

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